

Advanced SQL

NULL Functions

Exercise 5 – Practical SQL Worksheet

Instructions

- Write answers on paper with a pen.
- Draw tables of the final outcome.
- In the SELECT statement, choose relevant columns to display, if not specified.
- Scan the completed work into a PDF file and upload on GitHub
- Copy and email your GitHub link to: lerato@brightlearn.co.za
- Use the provided tables and write SQL queries to solve each question.
- Use NULL-handling functions (IFNULL(), ISNULL(), COALESCE(), etc.).
- For each question, match your output columns to the expected output format.

Table 1: Employees

employee_id	name	department	salary
1	Alice	HR	5000
2	Bob	IT	NULL
3	Charlie	NULL	7000
4	Dana	Finance	NULL

Q1. Show all employees with their salary. If salary is NULL, display 0.

Expected Columns: employee_id, name, salary_with_default

Q2. Show employee names with their department. If department is NULL, show "Not Assigned".

Expected Columns: employee_id, name, department_name

Table 2: Orders

order_id	customer_id	delivery_date
101	201	2024-12-01
102	202	NULL
103	NULL	2024-12-03

Q3. Find orders with NULL customer_id using ISNULL().

Expected Columns: order_id, customer_id

Q4. Show all orders. If delivery_date is NULL, show 'Pending'.

Expected Columns: order_id, customer_id, delivery_status

Table 3: Students

student_id	name	grade
1	Ethan	85
2	Maya	NULL
3	Olivia	90

Q5. Show all students and their grades. Replace NULL with 0.

Expected Columns: student_id, name, final_grade

Q6. Count students who haven't been graded.

Expected Columns: not_graded_count

Table 4: Products

product_id	name	price	discount
501	Keyboard	25	NULL
502	Mouse	15	5
503	Monitor	100	NULL

Q7. Show product name, price, and final price after discount (assume 0 if discount is NULL).

Expected Columns: product_id, name, final_price

Table 5: Customers

customer_id	name	email
1	Linda	NULL
2	Joseph	joseph@mail.com
3	Nia	NULL

Q8. Count how many customers have no email.

Expected Columns: missing_email_count

Q9. Show all customers with email. If NULL, display "No Email".

Expected Columns: customer_id, name, email_display

Table 6: Payments

payment_id	method	status
301	Credit	NULL
302	PayPal	Success
303	NULL	Failed

Q10. Show payment details with method replaced by "Unknown" if NULL.

Expected Columns: payment_id, method_display, status

Table 7: Inventory

item_id	item_name	quantity
1	Pen	NULL
2	Notebook	150
3	Eraser	NULL

Q11. Show items and their quantity (0 if NULL).

Expected Columns: item_id, item_name, quantity_checked

Table 8: Employees_Extra

emp_id	bonus	commission
1	NULL	300
2	100	NULL
3	NULL	NULL

Q12. Show employee ID and the first available value among bonus or commission.

Expected Columns: emp_id, first_available_reward

Table 9: Classes

class_id	subject	room
11	Math	NULL
12	Science	Lab A
13	English	NULL

Q13. Count classes that don't have a room assigned.

Expected Columns: no_room_count

Table 10: Attendance

student_id	date	status
1	2025-04-01	NULL
2	2025-04-01	Present
3	2025-04-01	Absent

Q14. Show attendance records with status. Replace NULL with "Not Marked".

Expected Columns: student_id, date, attendance_status

Table 11: Bank_Accounts

account_id	account_type	balance
A1	Savings	NULL
A2	Current	5000
A3	NULL	2000

Q15. Show account ID, account_type (or 'Unknown'), and balance (or 0).

Expected Columns: account_id, type_display, balance_checked

Table 12: Projects

project_id	title	start_date	end_date
1	Website Revamp	2025-01-10	NULL
2	Mobile App	NULL	2025-06-01
3	Data Migration	NULL	NULL

Q16. Show all projects with a start date. If start_date is NULL, display 'TBD'.

Expected Columns: project_id, title, start_display

Table 13: Reviews

review_id	product_id	comment	rating
1	501	Great product	4
2	502	NULL	NULL
3	503	Works fine	3

Q17. Display reviews showing comment (or 'No Comment') and rating (or 0).

Expected Columns: review_id, product_id, comment_display, rating_display

Table 14: Suppliers

supplier_id	name	phone	alt_phone
1	Global Goods	NULL	123456789
2	Best Supplies	987654321	NULL
3	ValueSource	NULL	NULL

Q18. Show the supplier contact number. Use COALESCE(phone, alt_phone, 'No Contact').

Expected Columns: supplier_id, name, contact_number

Table 15: User_Settings

user_id	theme	language	timezone
1	NULL	English	NULL
2	Dark	NULL	UTC+1
3	NULL	NULL	NULL

Q19. Show all users and their preferences. Replace all NULLs with defaults:

- Theme → "Light"
- Language → "English"
- Timezone → "UTC"

Expected Columns: user_id, theme_set, language_set, timezone_set

Table 16: Maintenance

record_id	machine_id	issue	technician
1	M101	Overheating	NULL
2	M102	NULL	NULL
3	M103	Jammed	Alex

Q20. Show maintenance log with:

- issue → default to "Unknown Issue"
- technician → default to "Not Assigned"

Expected Columns: record_id, machine_id, issue_log, technician_name